

UBUNTU ANALYTIQ

Data & AI Mastery Mentorship · 12-Week Program · Building for Us by Us

WEEK 01 · SESSION GUIDE

Foundations of Data That Matters

"You cannot build solutions without first understanding the data that shapes your world."

Dataset	Wk 01 – Supply Chain Dataset.xlsx (4,500 rows)
Tools	Python 3.11, Pandas, Matplotlib, Seaborn, Jupyter Notebook
Duration	3–4 hours self-study + 1-hour check-in
Deliverable	Documented EDA notebook + 1-page data quality report

Session Overview

This is where your journey begins. Before you automate, predict, or deploy - you must first read data like a professional. This session grounds you in the Nairobi retail supply chain context: real stockout patterns, real supplier delays, real gaps that cost businesses money every week. Your job is to find **the story the data is trying to tell**.

Dataset Context

The dataset simulates 12 months of daily stock movements across a Nairobi distributor network covering 15 product lines from 10 Kenyan suppliers across 7 Nairobi regions. It includes: units ordered vs received (supplier fulfilment gaps), units sold vs stock on hand (demand velocity), lead times in days (supplier reliability), stockout flags (critical events), and regional breakdown.

💡 Business Context: In Kenyan retail, stockouts of staple goods like unga, cooking oil, and milk can cost a distributor 15–25% of weekly revenue. Understanding WHERE and WHY stockouts happen is the first step to automating their prevention (which you will do in Week 8).

Step-by-Step Walkthrough

Phase 1: Environment Setup

Step	Action	Expected Output
1A	Create a new Jupyter notebook named 'week01_supply_chain_eda.ipynb'. Import: pandas as pd, matplotlib.pyplot as plt, seaborn as sns, numpy as np. Set seaborn style to 'darkgrid'.	Notebook running, libraries loaded without errors

1B	Load the dataset: <code>df = pd.read_excel('week01-nairobi-supply-chain.xlsx', sheet_name='Supply_Chain_Logs', header=4)</code> . Check <code>df.head()</code> , <code>df.shape</code> , <code>df.dtypes</code> , <code>df.info()</code>	DataFrame with 4,500 rows × 12 columns confirmed
1C	Parse the Date column: <code>df['Date'] = pd.to_datetime(df['Date'])</code> . Extract Year, Month, Quarter, DayOfWeek as new columns.	Date column is datetime type; new time columns visible in <code>df.head()</code>

Phase 2: Data Quality Assessment

A data quality report is not optional - it is the first professional deliverable. Never present analysis on unchecked data.

Step	Action	Expected Output
2A	Check missing values: <code>df.isnull().sum()</code> . Calculate missing percentage per column. Print a formatted report.	Table showing column name, missing count, missing %
2B	Check for logical inconsistencies: <code>units_received</code> should never exceed <code>units_ordered</code> . Flag rows where <code>Units_Received > Units_Ordered</code> .	Count of anomaly rows (expect 0–50 in this dataset)
2C	Check for duplicate TransactionIDs: <code>df.duplicated('TransactionID').sum()</code> . If any found, investigate and document.	Duplicate count printed; if >0, describe the pattern
2D	Calculate a Data Quality Score: $(1 - \text{total_issues} / \text{total_cells}) \times 100$. Print: 'Data Quality Score: XX.X%'	Single quality score between 85–100% expected

Phase 3: Exploratory Analysis — The 5 Questions (60 minutes)

Every EDA should answer business questions, not just produce charts. Here are your 5 questions for this dataset:

› Q1: Which product categories have the highest stockout rate?

```
stockout_rate = df.groupby('Category')['Stockout_Flag'].apply(lambda x:
(x=='Yes').mean()).sort_values(ascending=False) print(stockout_rate)
```

› Q2: Which suppliers have the longest average lead times?

```
lead_time = df.groupby('Supplier')['Lead_Time_Days'].mean().sort_values(ascending=False)
lead_time.plot(kind='barh', title='Avg Lead Time by Supplier', color='#00C851')
```

› Q3: Is there a seasonal pattern in stockouts? (Month-over-month)

```
monthly_stockouts = df[df['Stockout_Flag']=='Yes'].groupby(['Year', 'Month']).size().reset_index(name='Count')
plt.plot(monthly_stockouts['Count'])
```

› Q4: Which regions have the worst stock availability?

```
region_stockout = df.groupby('Region')['Stockout_Flag'].apply(lambda x:
(x=='Yes').sum()).sort_values(ascending=False)
```

› Q5: What is the supplier fulfillment rate? (Units_Received / Units_Ordered)

```
df['Fulfillment_Rate'] = df['Units_Received'] / df['Units_Ordered']
fulfillment = df.groupby('Supplier')['Fulfillment_Rate'].mean().sort_values()
```

Phase 4: Visualisation Dashboard

Create a 2×3 figure with `plt.subplots(2,3,figsize=(15,10))`. Place your 5 charts (one per question) plus a summary heatmap. Use colour #00C851 for positive metrics, #E63946 for negative/risk metrics. Title each chart clearly. Save as 'week01_dashboard.png'.

Weekly Deliverable

Submit the following to the Academy portal by your check-in session:
1. week01_supply_chain_eda.ipynb — fully run notebook with all cells executed and outputs visible
2. week01_dashboard.png — the 2×3 chart dashboard
3. week01_data_quality_report.md — a 1-page markdown report containing: Data Quality Score, top 3 data issues found, top 3 business insights, and one question the data cannot answer (and why).

Assessment Rubric

Criterion	What We Look For	Points
Data Loading & Cleaning	Dataset loaded correctly, Date parsed, quality score calculated	20
5 Business Questions	All 5 questions answered with correct code and interpretation	35
Visualisation Quality	Charts are titled, labelled, readable, and use appropriate chart types	20
Data Quality Report	All 3 required sections present; insights are specific and business-relevant	25
TOTAL		100